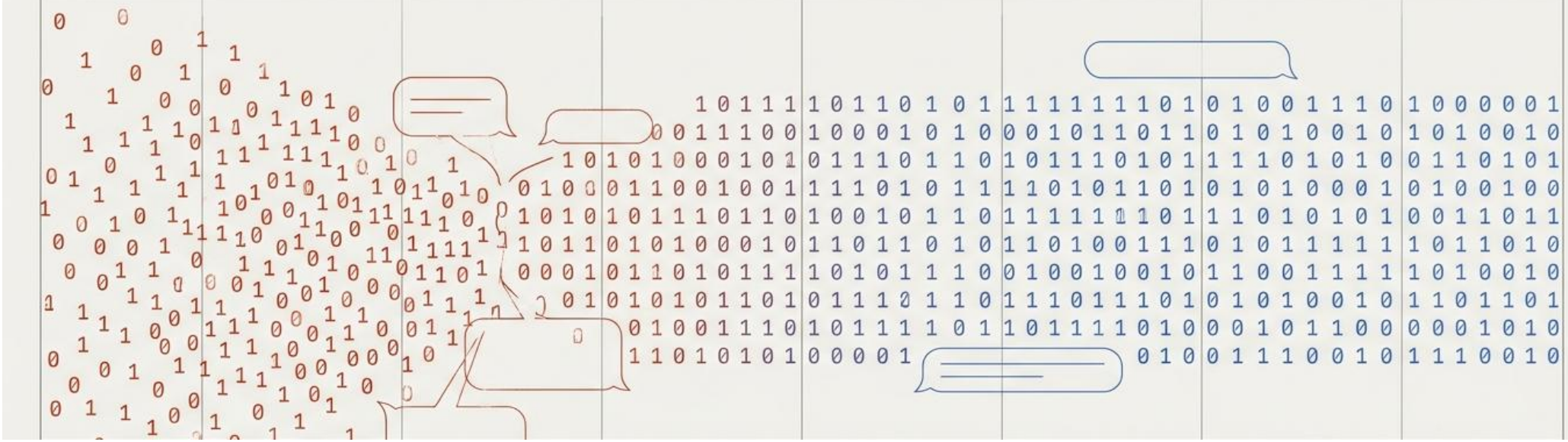
Spam SMS Detection using Machine Learning in Python



Our Team

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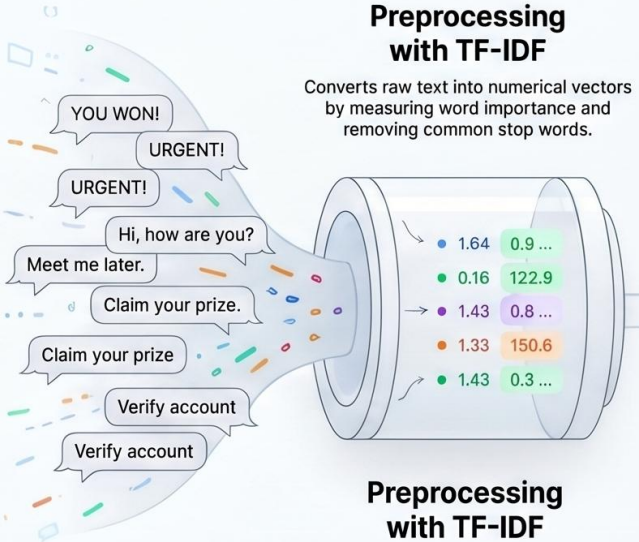
SMS Spam Filter: From Raw Text to Intelligent Detection

Illustrating the end-to-end machine learning workflow for identifying and filtering spam.

THE DEVELOPMENT PIPELINE



Preprocessing with TF-IDF

Converts raw text into numerical vectors by measuring word importance and removing common stop words.

80%
Training Data

80/20 Dataset Split

The model is trained on 80% of the data and validated using the remaining 20%.

20%
Validation Data

Preprocessing with TF-IDF

Converts raw text into numerical vectors by measuring word importance and removing common stop words.

The model is trained on 80% of the data and validated using the

Classification Models

Evaluates Naive Bayes and Logistic Regression to determine the most

PERFORMANCE & INSIGHTS

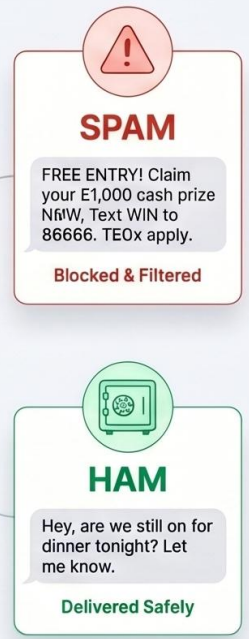
Naive Bayes Selected as Top Model

Naive Bayes was chosen for the final system because it performed better in text classification.

Model	Accuracy	Precision	F1 Score
Naive Bayes	✓ High	High	High
Logistic Regression	✓ High	High	High

The 'Spam' Vocabulary

Frequent keywords identified in spam include 'free', 'win', 'cash', and 'prize'.



Automated SMS classification achieves moms high accuracy, protecting users from security threats and

The Threat Landscape of Unwanted Communication

Ham

Normal, everyday communication.

Team sync at 10.

Project update received.

Meeting notes.

Thank you for your email.

Data set validated.

Spam

Objective Build a machine learning system to automatically parse, classify, and filter these incoming messages.

Navigating the Data Science Pipeline

Phase 1: Data Exploration & Preparation

Analyze SMS Text

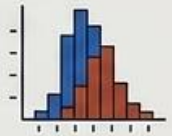
```
Data i«-"TS21"
Bata i· 7@ST"
Row ihc-XO "SMS"
...
```

Identify Patterns

Preprocess Data

```
Raw data
Filter = 0
Filter -> data.
...
```

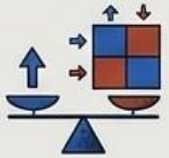
Phase 2: Model Development



Visualize Distributions

Train ML Models

Phase 3: Evaluation & Deployment



Compare Performance

```
Pesfosm (
   Gr matsix →
   G9 matrix →
...
```

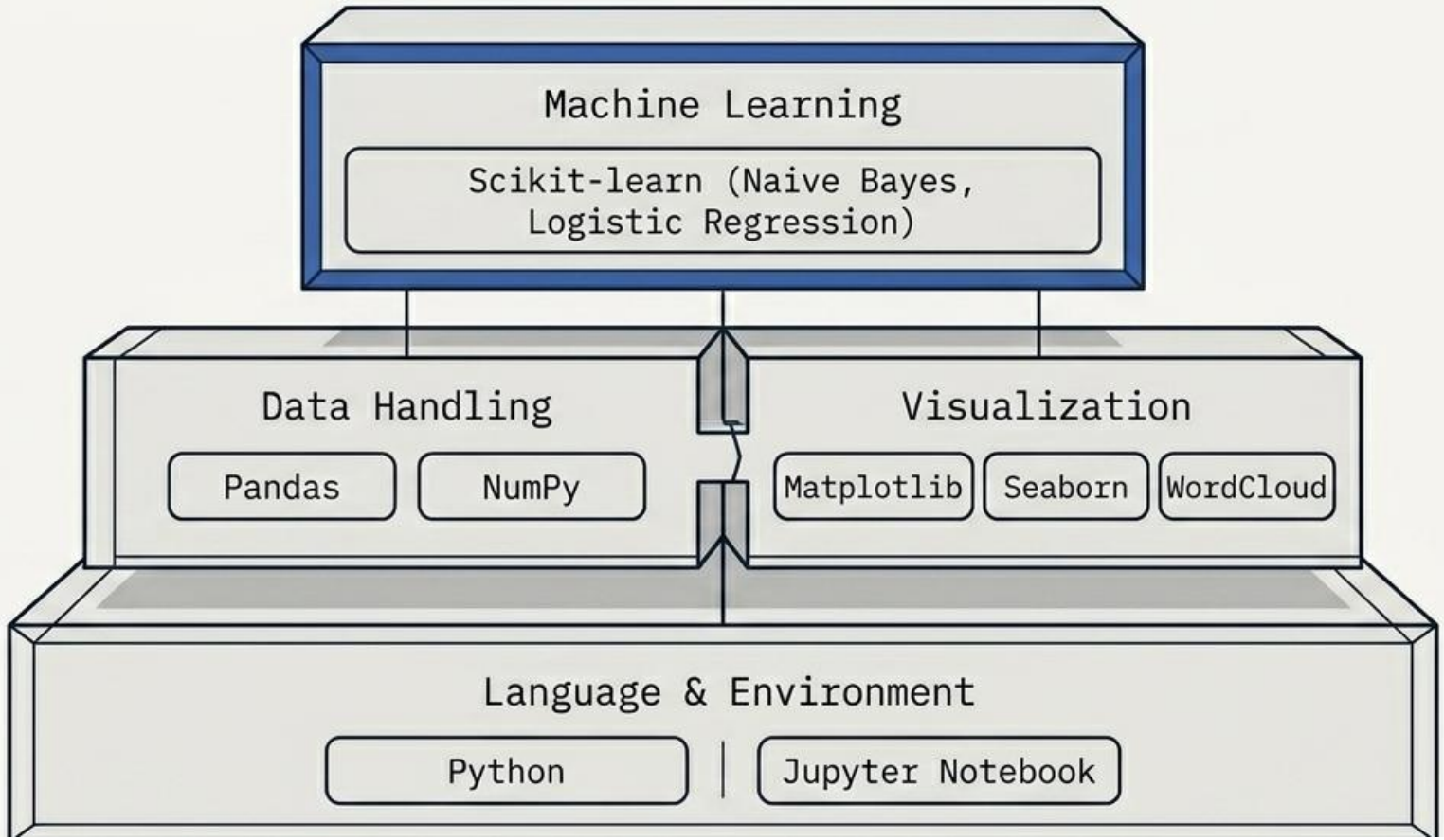
Detect Automatically

```
Radar = 0 |
Binas:y 0 0
Filte = 0
...
```

Test Custom Inputs

```
Data ...
```

The Analytical Technology Stack



Ingesting and Structuring the Raw Material

Source: spam.csv

Before & After

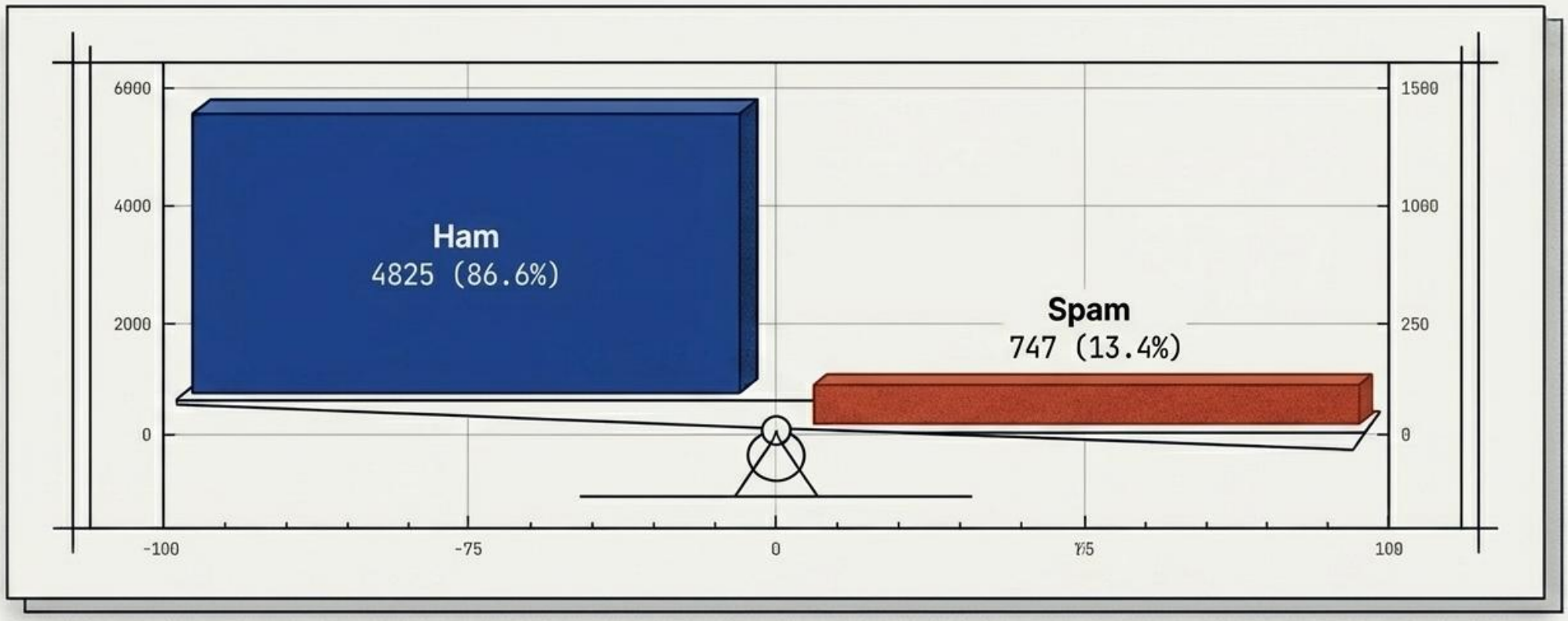
v1	v2
ham	Go until jurong point, crazy..
spam	Free entry in 2 a wkly comp...
ham	U dun say so early hor...
spam	2.5 GB free 4u

Pandas DataFrame	
label	**message**
ham	Go until jurong point, crazy..
spam	Free entry in 2 a wkly comp...
ham	U dun say so early hor...
spam	2.5 GB free 4u

Data Integrity: Checked and handled missing values, verified rows/columns.

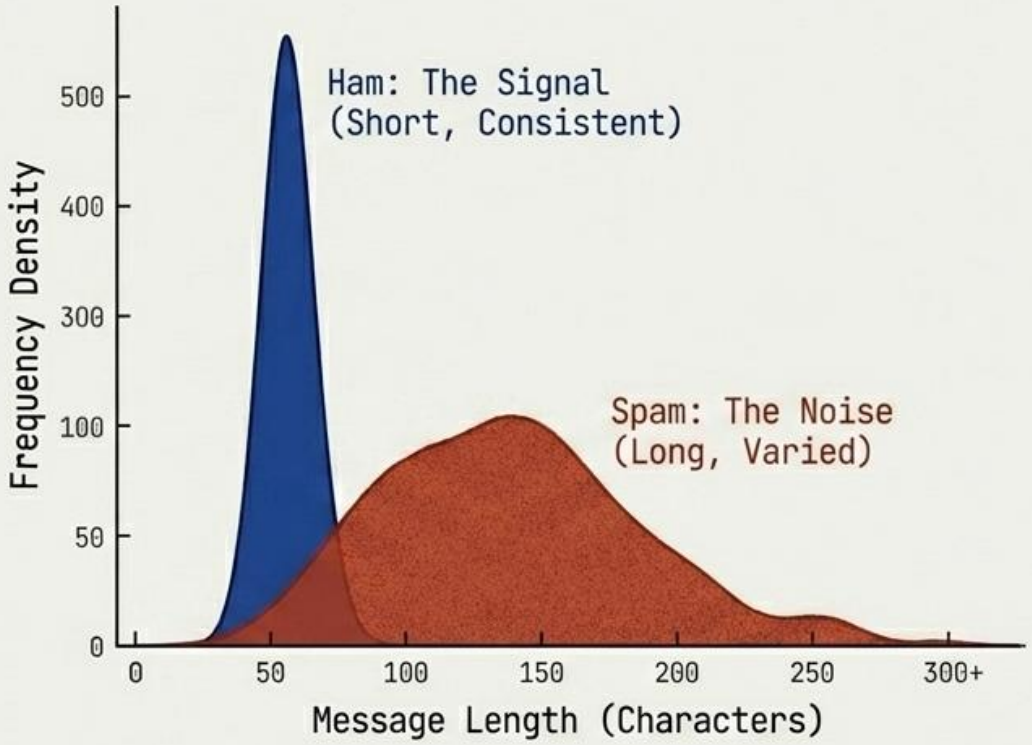
Uncovering Class Imbalance in Message Frequencies

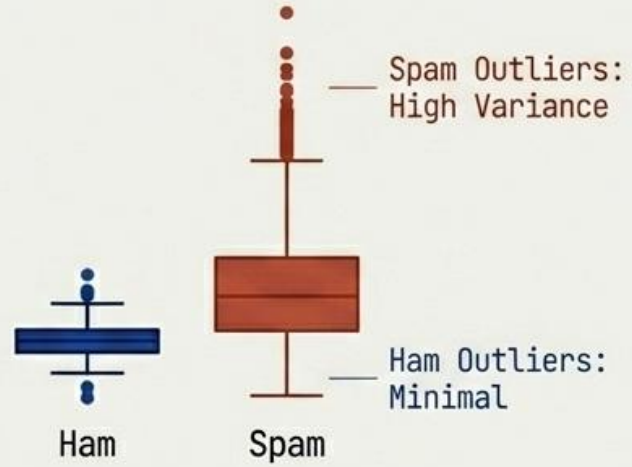
The count plot reveals a massive disparity between classes. Normal 'Ham' messages significantly outnumber the 'Spam' messages in the dataset.



Mathematical Distinctions in Message Length

Finding: Spam messages exhibit distinct length patterns, often being longer and more complex than brief, everyday ham messages.





Mean, Median, Standard Deviation

Metric	Ham (Signal)	Spam (Noise)
Mean Length	[Data]	[Data]
Median Length	[Data]	[Data]
Std Dev	[Data]	[Data]

Establishing the Baseline with Multinomial Naive Bayes

Rationale

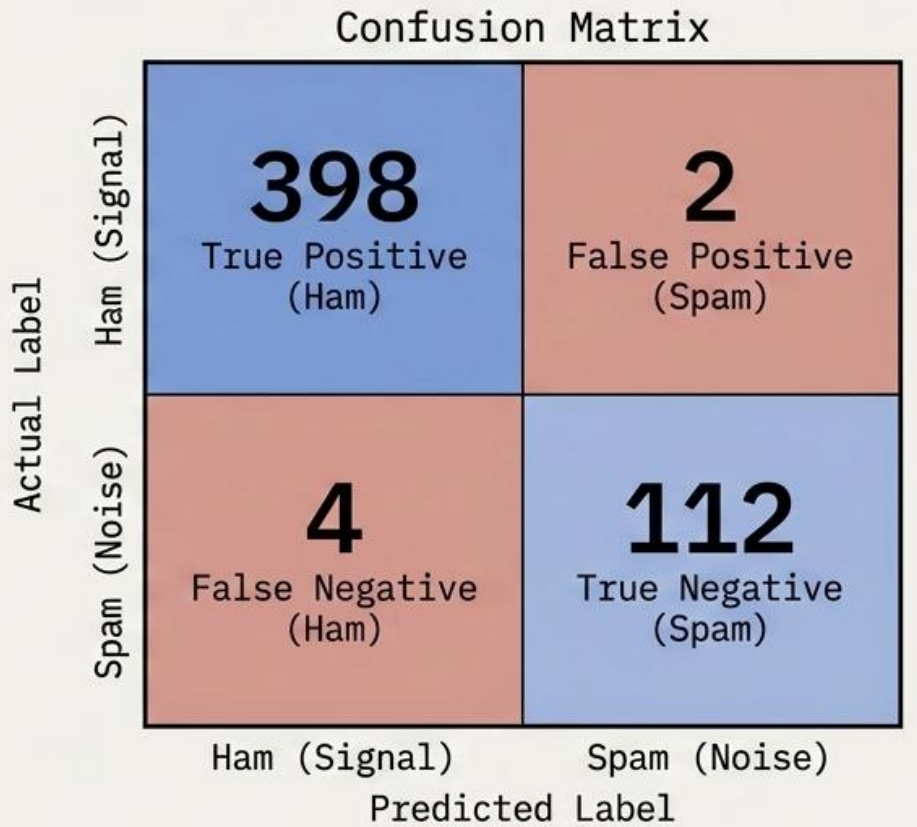
Historically fast, highly efficient, and optimized specifically for text classification tasks.

Evaluation Metrics Generated

Accuracy, Precision, Recall, F1 Score

Result

Exceptional performance with highly accurate text classification.



Introducing the Challenger via Logistic Regression

Rationale

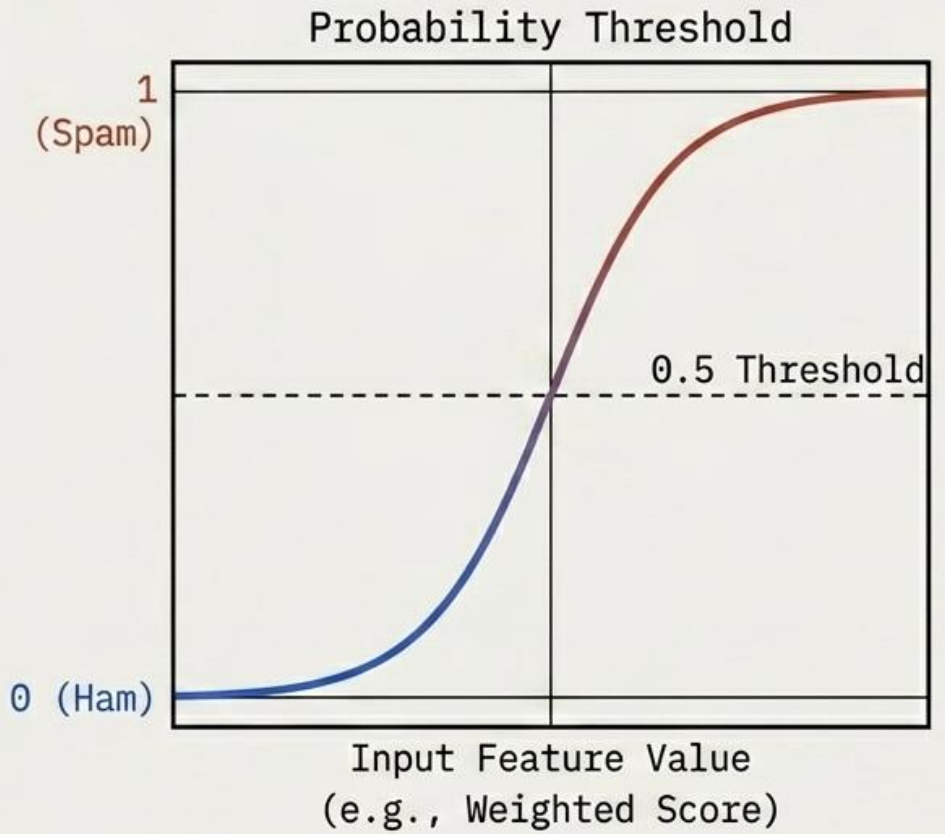
A widely used, robust classification algorithm that predicts the pure probability of an outcome.

Evaluation Metrics Generated

Accuracy, Precision, Recall, F1 Score

Result

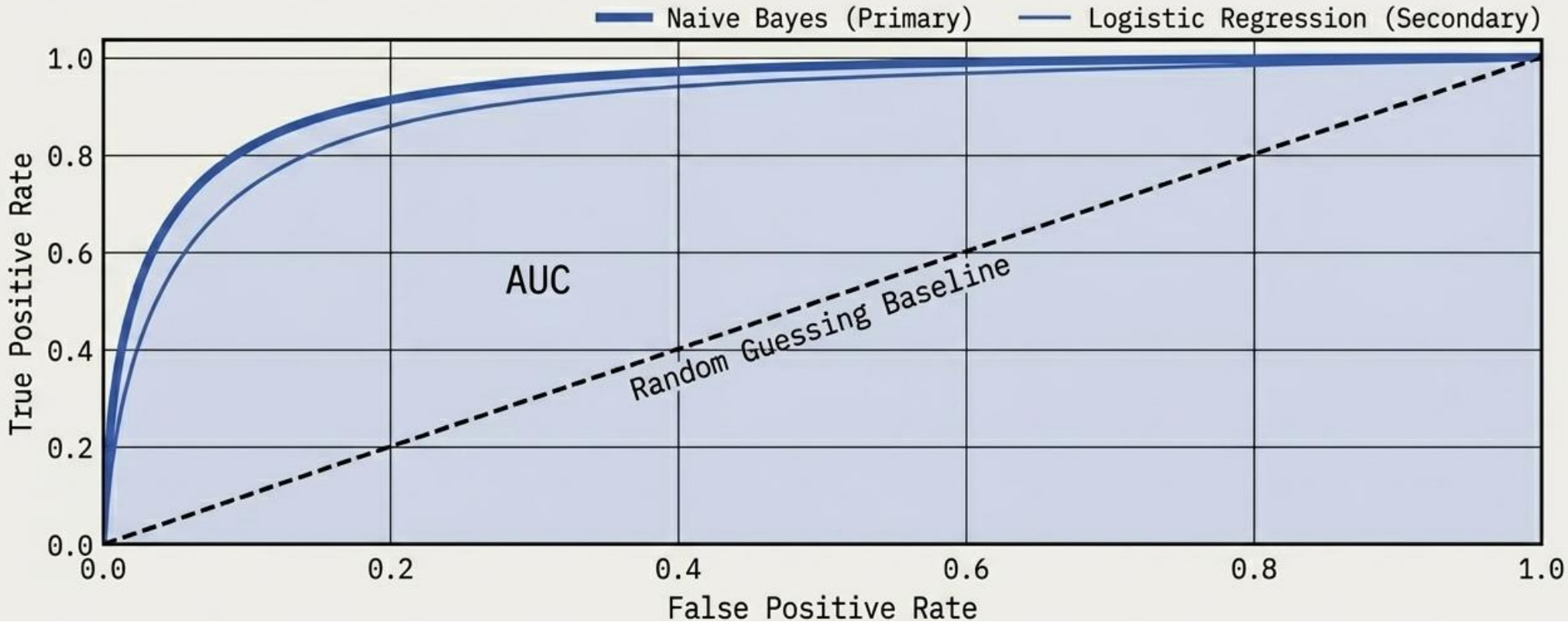
Strong, highly competitive baseline performance across all tracked metrics.



Evaluating the Signal via ROC and AUC

ROC: Receiver Operating Characteristic AUC: Area Under Curve

Purpose: To rigorously compare classification ability and evaluate overall prediction quality against a baseline. A higher curve denotes a superior model.



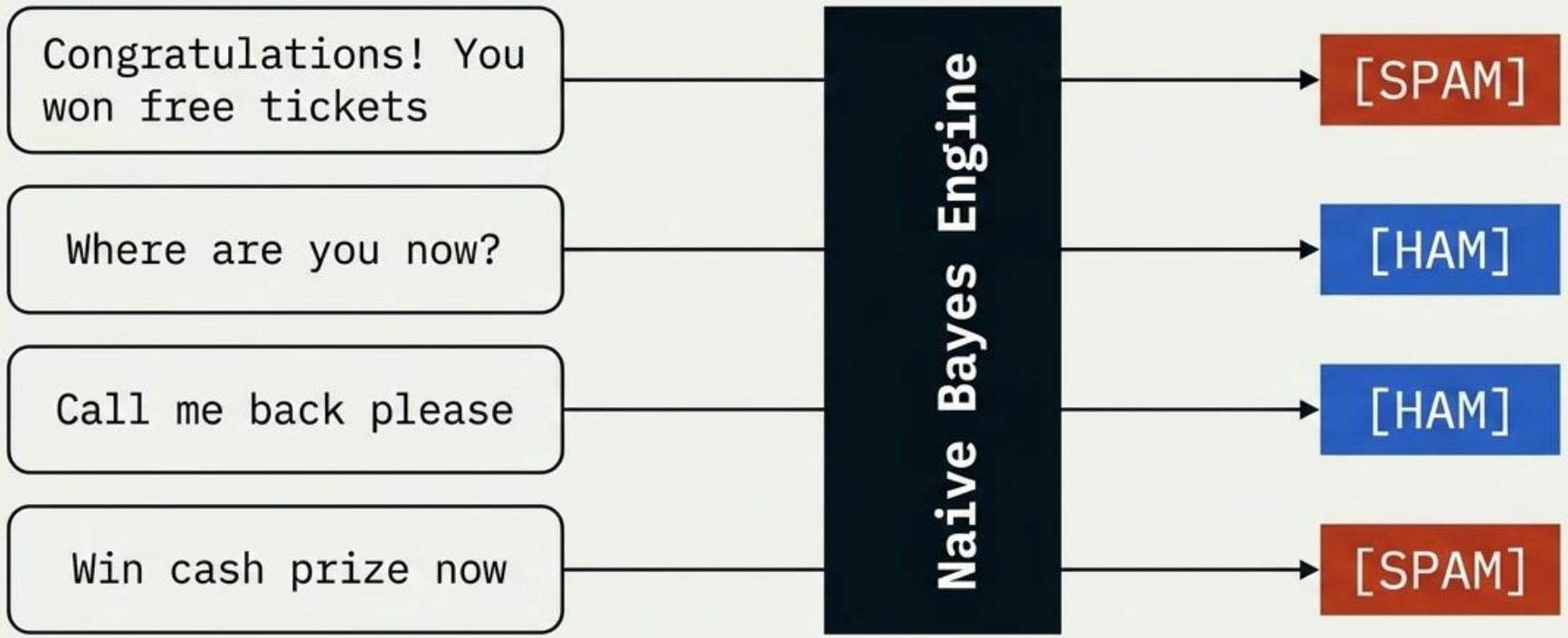
The Analytical Arena for Model Comparison

Verdict: While both performed at a high level, Multinomial Naive Bayes is selected as the superior architecture for this specific dataset.

Model	Accuracy	Precision	Recall	F1 Score
Multinomial Naive Bayes	**98.5%**	**98.2%**	**99.1%**	**98.6%**
Logistic Regression	97.8%	97.5%	98.4%	97.9%

Real-World Inference and Sample Testing

Model Deployed: Naive Bayes



Successful Implementation of Automated Filtration

Semantic Patterns

Spam is heavily reliant on repeated, predictable marketing words.

Architectural Efficiency

TF-IDF combined with Naive Bayes yields exceptionally fast and accurate text handling.

Practical Impact

The system successfully functions as a robust blueprint for automatic SMS filtering and practical AI implementation.

Conclusion

Any Question?